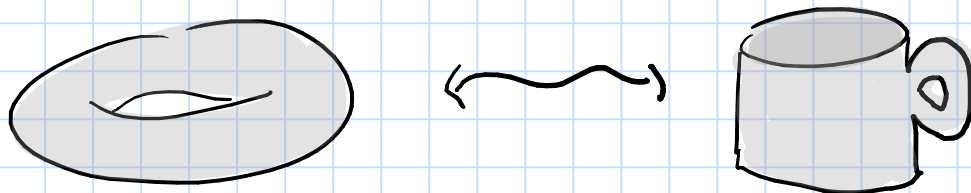


MOTIVATION

We said that broadly speaking **TOPOLOGY** studies shapes through their continuous deformations.



Before giving the abstract definition of a topological space, we will try to answer the following question:

WHAT DO WE NEED TO DEFINE CONTINUITY?

Let's go back to what we know from undergrad. real analysis.

$f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous

$\Leftrightarrow \forall x \in \mathbb{R}$ f is continuous at x

$\Leftrightarrow \forall x \in \mathbb{R}, \forall \epsilon > 0, \exists \delta > 0$ st.

$$\underbrace{|f(y) - f(x)|}_{\text{dist}(f(y), f(x))} < \epsilon \quad \forall y \in \mathbb{R} \text{ st. } \underbrace{|y - x|}_{\text{dist}(y, x)} < \delta$$

It seems that we need a notion of DISTANCE.

A reasonable distance should satisfy appropriate axioms, which lead to the definition of metric space.

Definition. Let X be a nonempty set. A **metric** (or **distance**) on X is a function $d : X \times X \rightarrow [0, \infty[$ such that

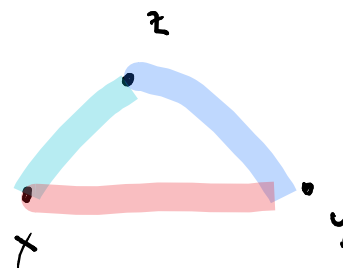
i) $d(x, y) = 0$ if and only if $x = y$

ii) $d(x, y) = d(y, x)$ **SYMMETRY**

iii) $d(x, y) \leq d(x, z) + d(z, y)$ **TRIANGLE INEQUALITY**

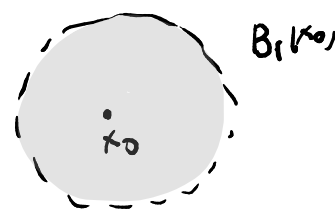
for all $x, y, z \in X$.

The pair (X, d) is called a **metric space**.



Definition. Let (X, d) be a metric space, $x_0 \in X$. We define the **open ball** centered at x_0 with radius $r > 0$ as

$$B_r(x_0) = \{x \in X \mid d(x, x_0) < r\}.$$

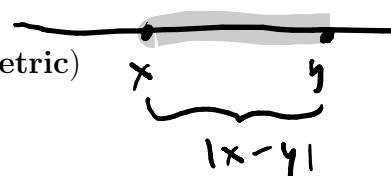


Example 1 Consider

$$d(x, y) = |x - y|$$

for all $x, y \in \mathbb{R}$. Then $(\mathbb{R}, d)$ is a metric space.

(The distance d above is the standard metric on $\mathbb{R}$, called the **Euclidean metric**)



proof let $x, y, z \in \mathbb{R}$. We have

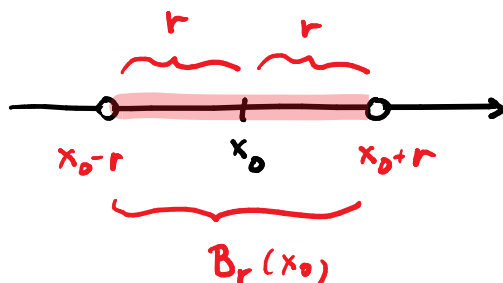
1. $d(x, y) = 0 \Leftrightarrow |x - y| = 0 \Leftrightarrow x - y = 0 \Leftrightarrow x = y$

2. $d(x, y) = |x - y| = |y - x| = d(y, x)$

3. $d(x, y) = |x - y| = |x - z + z - y| \leq |x - z| + |z - y| = d(x, z) + d(z, y)$

A ball in this metric space is an interval. If $x_0 \in \mathbb{R}$, $r > 0$, then

$$B_r(x_0) = \{x \in \mathbb{R} \mid |x - x_0| < r\} = \{x \in \mathbb{R} \mid x_0 - r < x < x_0 + r\} =]x_0 - r, x_0 + r[$$



Example 2 Let X be a nonempty set and define

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

for all $x, y \in X$. Then d is a metric on X , called the **discrete metric**.

• What do balls look like?

Let $x_0 \in X, r > 0$.

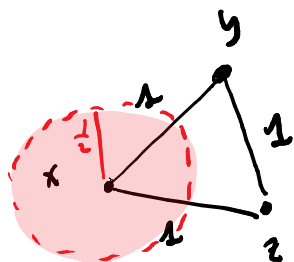
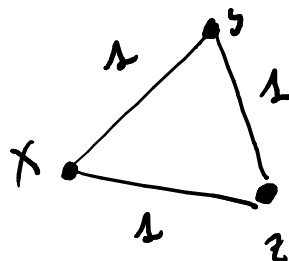
Then $B_r(x_0) = \{x_0\}$ if $r \leq 1$

and $B_r(x_0) = X$ if $r > 1$

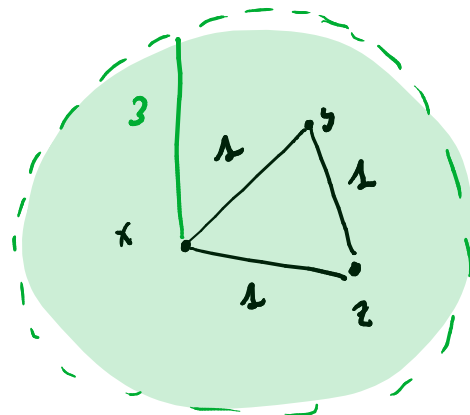
so Balls are either singletons or the whole space!

If X has 3 elements $X = \{x, y, z\}$ we can visualize

the discrete metric as



$B_{\frac{1}{2}}(x)$ contains
only x !



$B_3(x)$
contains
all points!

Example 3 For all $x, y \in \mathbb{R}^n$ define

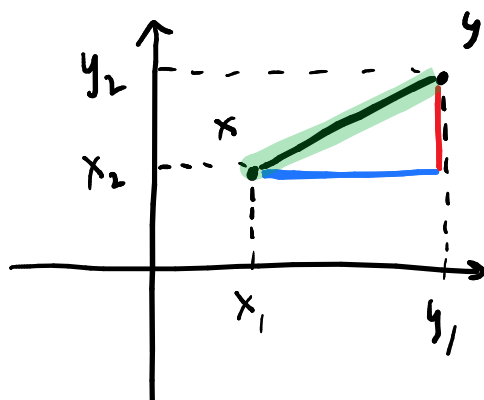
$$d_2(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{\frac{1}{2}} \quad d_1(x, y) = \sum_{i=1}^n |x_i - y_i| \quad d_\infty(x, y) = \max_{i=1, \dots, n} |x_i - y_i|$$

Then, d_2, d_1, d_∞ are three distinct metrics on $\mathbb{R}^n$.

Let's do some visuals in $\mathbb{R}^2$.

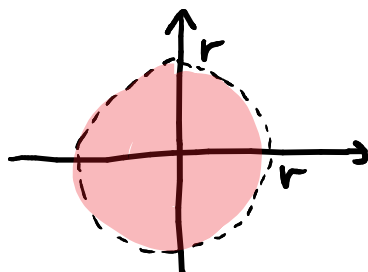
Let $x = (x_1, x_2)$ $y = (y_1, y_2) \in \mathbb{R}^2$

- look at d_2 first. This is the Euclidean metric.



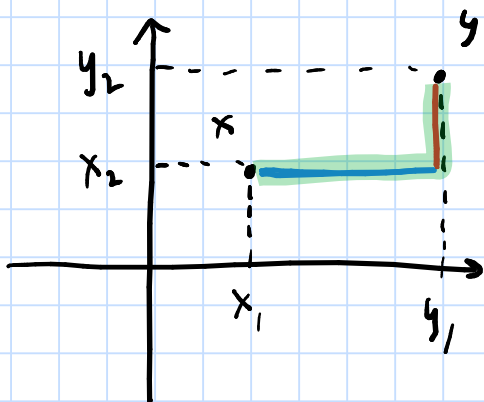
$$d_2(x, y) = \left(\underbrace{|x_1 - y_1|^2}_{\text{blue}} + \underbrace{|x_2 - y_2|^2}_{\text{red}} \right)^{\frac{1}{2}}$$

= length of segment

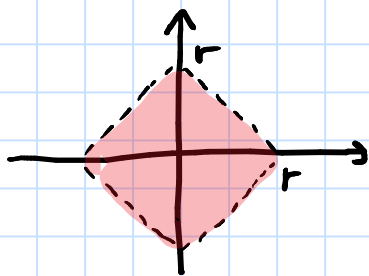


$$B_r(0) = \{ (x_1, x_2) \in \mathbb{R}^2 \mid \sqrt{x_1^2 + x_2^2} < r \}$$

- Now look at d_1 . This is also called Taxicab (or Manhattan) metric

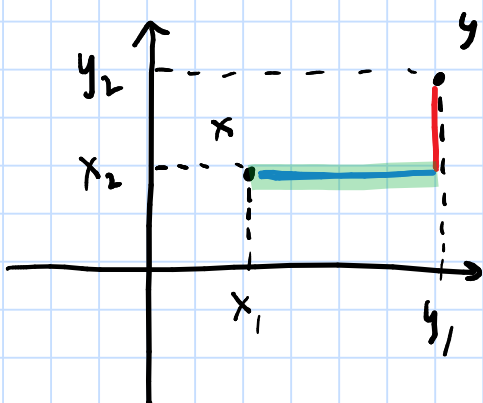


$$d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

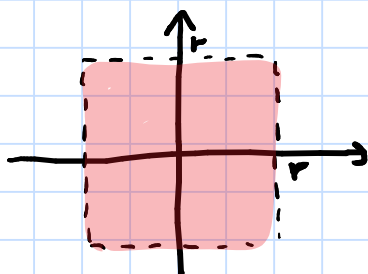


$$B_r(0) = \{ (x_1, x_2) \in \mathbb{R}^2 \mid |x_1| + |x_2| < r \}$$

- Lastly, look at d_∞ .



$$d_\infty(x, y) = \max \{ |x_1 - y_1|, |x_2 - y_2| \}$$



$$B_r(0) = \{ (x_1, x_2) \in \mathbb{R}^2 \mid \max\{|x_1|, |x_2|\} < r \}$$

Example 4 For all $f, g \in C[a, b]$ define

$$d_{\infty}(f, g) = \sup_{x \in [a, b]} |f(x) - g(x)| \quad \text{and} \quad d_1(f, g) = \int_a^b |f(x) - g(x)| dx.$$

Then d_1 and d_{∞} are two metrics on $C[a, b]$.

